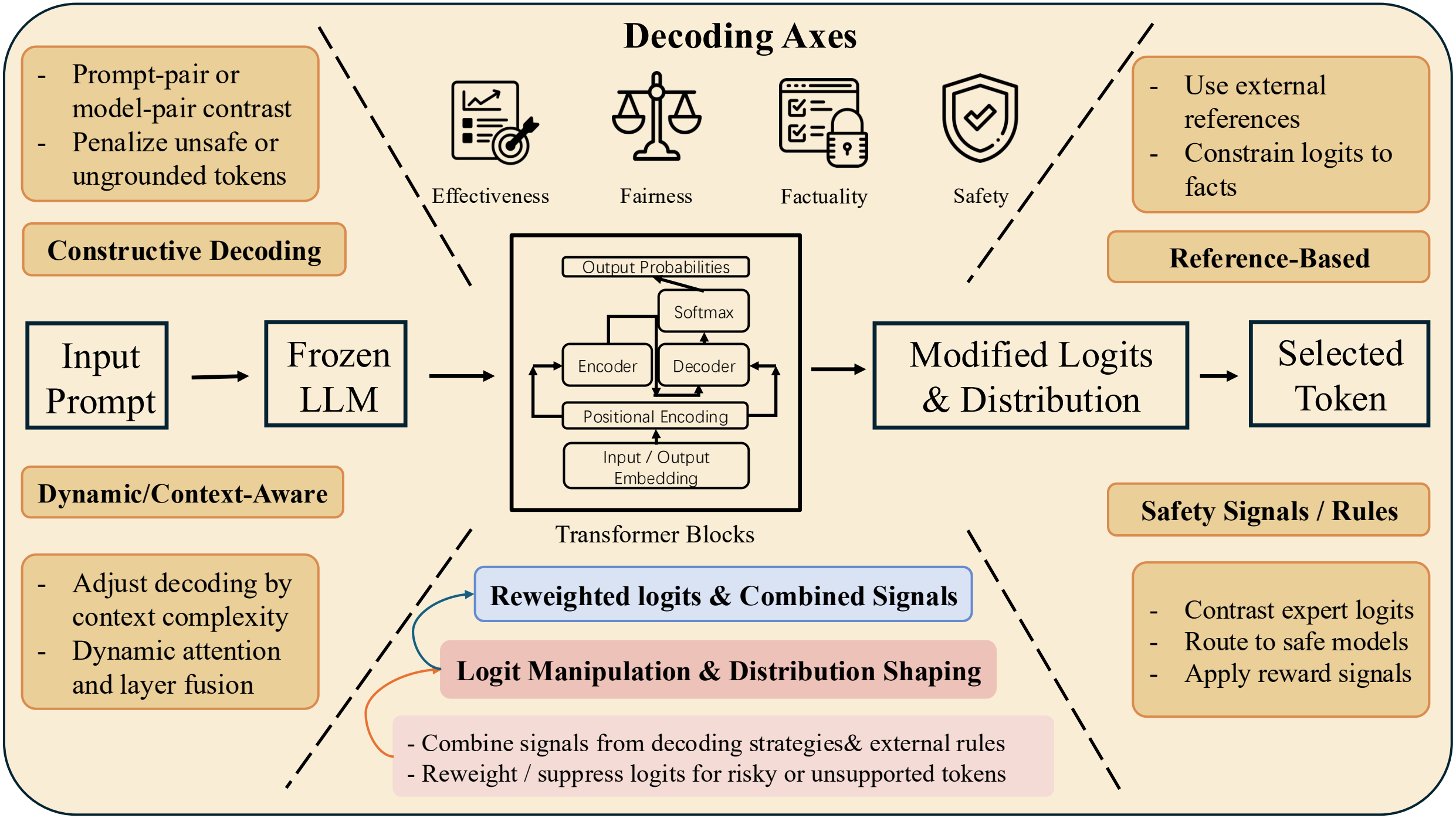

Decoding Axes
Effectiveness
Fairness
Factuality
Safety
- Prompt-pair or model-pair contrast
- Penalize unsafe or ungrounded tokens
Constructive Decoding
- Use external references
- Constrain logits to facts
Reference-Based
Input Prompt
Frozen LLM
Output Probabilities
Softmax
Encoder
Decoder
Positional Encoding
Input / Output Embedding
Transformer Blocks
Modified Logits & Distribution
Selected Token
Dynamic/Context-Aware
- Adjust decoding by context complexity
- Dynamic attention and layer fusion
Safety Signals / Rules
- Contrast expert logits
- Route to safe models
- Apply reward signals
Reweighted logits & Combined Signals
Logit Manipulation & Distribution Shaping
- Combine signals from decoding strategies& external rules
- Reweight / suppress logits for risky or unsupported tokens